

Cranial Chaos Gameplay: Illustration

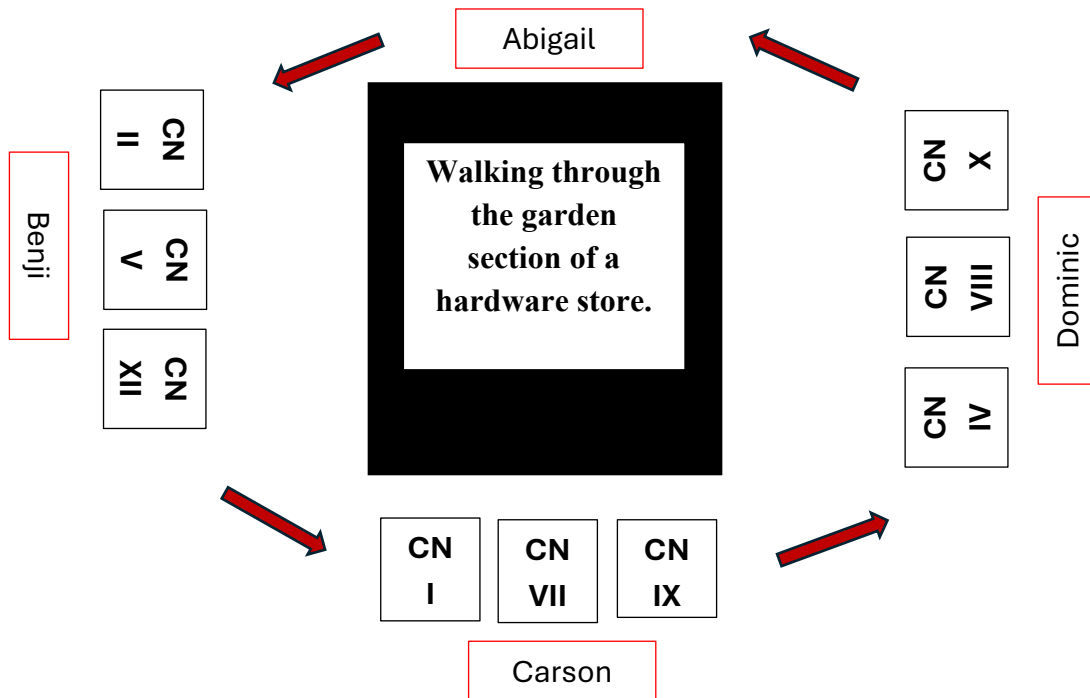
This is an illustration of the game play for single players. For ease, we will use four players, each with three of the cranial nerves represented in their hands, and four rounds of play to cover an example for each nerve. This can also be used for teams, with one player representing each team.

Players: Abigail, Benji, Carson, and Dominic

Each draw three cards and since Abigail was the last player to go on rotation at the hospital, she will be the judge in round one.

Round One:

Abigail turns over the top card of the Axon Deck to read. The remaining three players for this round will choose the best answer they have available in their hands and justify their choice with an explanation. Once all players laid their choice face down, they may reveal and give their justification to the judge starting with the player to the judge's left.



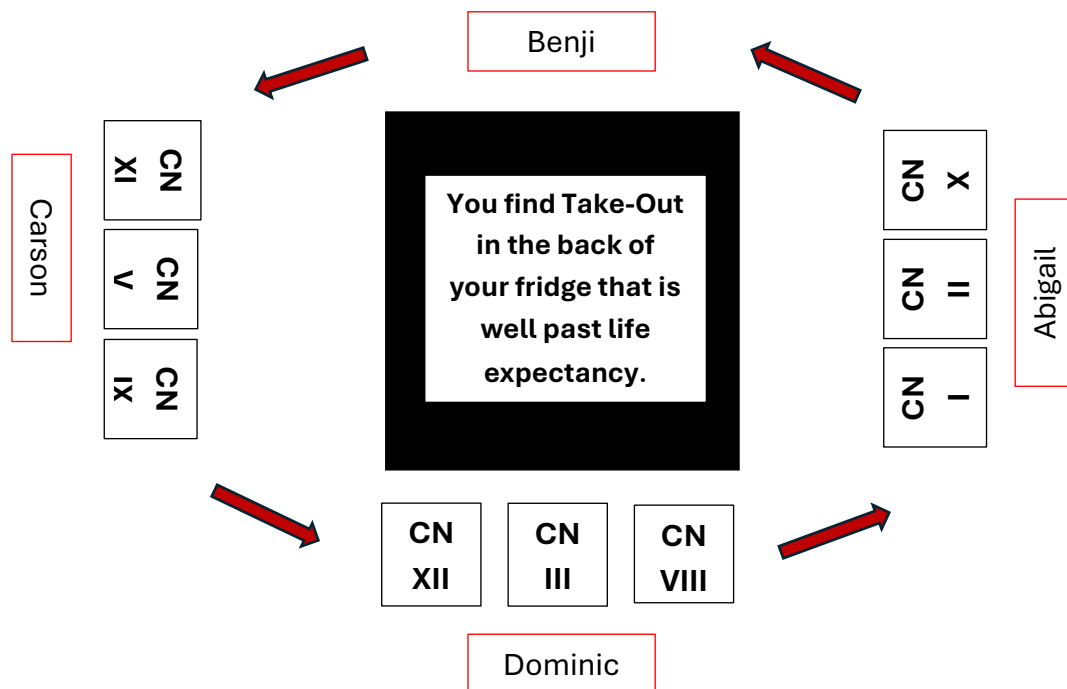
- **Benji** reveals his card: “I have CN II – the Optic Nerve. I chose it because I want to be able to see and use my vision while I’m shopping so I can pick the items I came in for.”
- **Carson** is next: “I picked CN VII – the Facial Nerve, because I love plants so when I walk through the garden section I want to be able to smile using the facial nerve for my superficial facial muscles like Zygomaticus Major.”

- **Dominic** finishes the round: “When I go shopping, my eyes are always moving. So, I chose CN IV – the Trochlear Nerve, because I’m hoping to catch a deal out of the corner of my eye with my superior oblique muscle.”

While all their answers are accurate, Abigail liked Carson’s reasoning the best and gives them the prompt Axon card to award them the point for the round.

Round Two:

Benji is now the judge and turns over the next Axon card for the prompt.

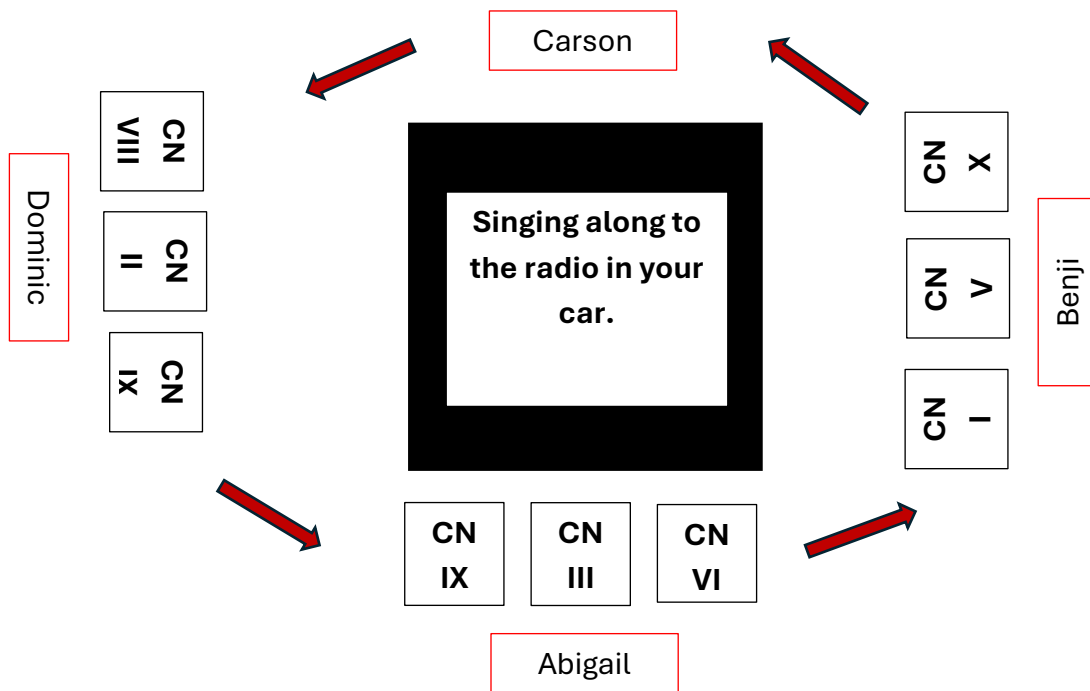


- **Carson** starts this round: “I’m playing CN XI – the Accessory Nerve. I know this innervates the sternocleidomastoid muscle and when I go to dump the bad food in the trash, I want to be able to Turn my head far, Far away from that.”
- **Dominic** follows: “I agree! I think this is going to be really gross so I’m using my CN XII – Hypoglossal Nerve. I’m going to make a disgusted face and stick out my tongue when I clean out the fridge.”
- **Abigail** is last: “Well, I think I need to double check how bad it got, maybe it’s still good! I’ll use CN I – Olfactory Nerve, to smell the food before throwing it out.”

Benji agreed that no matter how long the food had been in his fridge, he probably couldn’t help but sniff test it first too, so Abigail gained the point for round two.

Round Three:

Carson is the judge and reveals the prompt.



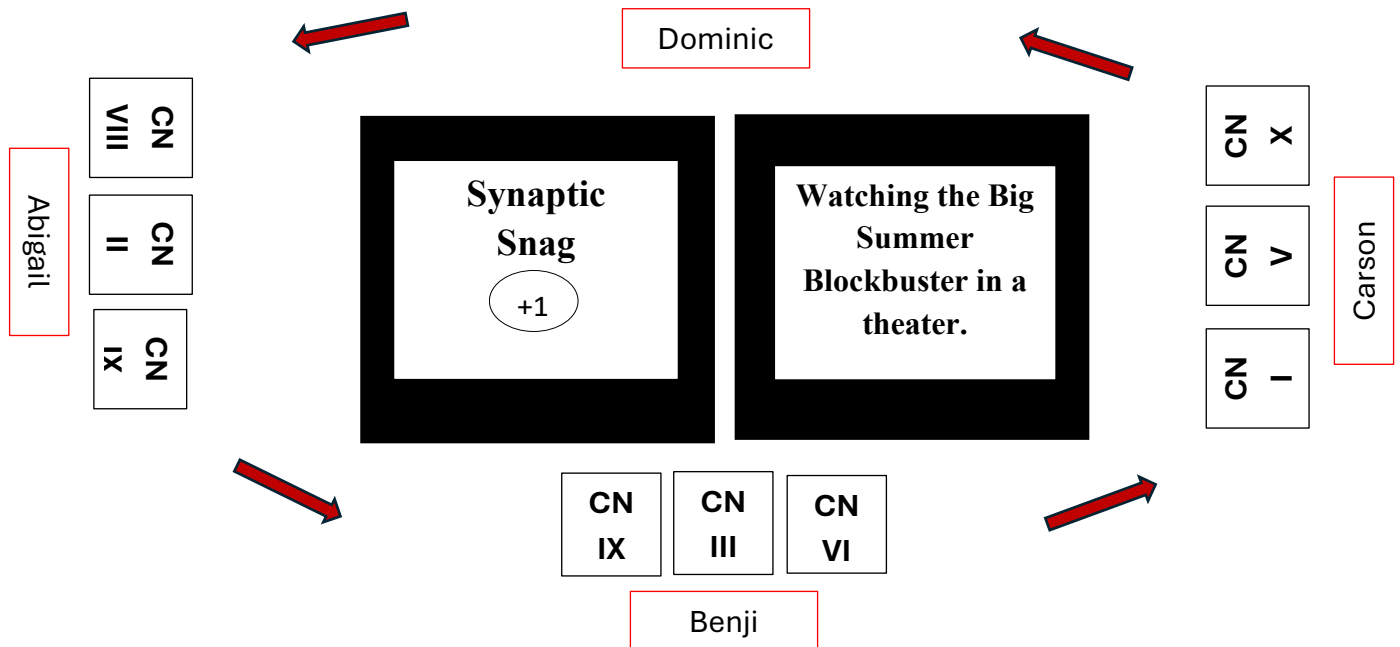
- **Dominic** begins: "Perfect! I have CN VIII – Vestibulocochlear Nerve. This way I can hear the radio and make sure I'm in sync with the music."
- **Abigail** goes next: "Well I'm not a good singer and knowing my luck, I would take a sip of water or something between songs and end up activating my gag reflex and coughing. So, I am going to play CN IX – the Glossopharyngeal Nerve, since that is the afferent part of that reflex."
- **Benji** ends the round: "I love singing, but without your vocal cords moving, you can't! I'm going to use CN X – the Vagus Nerve, because the recurrent laryngeal branch of vagus nerve innervates the intrinsic muscles of the vocal cords so I can sing in tune."

Carson awards the round to Benji for his point about the recurrent nerve.

Round Four:

Dominic turns over the top card, but rather than a prompt, a bonus card is revealed, “Synaptic Snag +1”. Dominic knows to turn over a second Axon card for a prompt.

The bonus card being turned over first indicates that for this round, the players must include in their answers where the nerve they are discussing synapses, should they add the accurate information and win the round, they will receive Two points this time.



- **Abigail** starts the last round: “I’m playing CN V – the Trigeminal Nerve, which does many things but I’m going to focus on V3 – the Mandibular branch. This branch innervates the muscles of mastication, and I can’t see a movie without some popcorn! Oh, and because that’s motor, it should synapse on the target muscles.”
- **Benji’s** turn: “I also love popcorn, but I like stealing it out of my partner’s bucket when they are watching movies. So, I’m going to use CN VI – Abducens Nerve, that way I can use my lateral rectus muscle to peek at them beside me, so I know when to strike. That’s also purely motor so it will synapse right on the lateral rectus muscle.”
- **Carson** will end the game: “I’m picking CN III – Oculomotor. Most of my eye movement to follow the screen will be done with this one, and I have a bad habit of falling asleep in theaters so I’m going to use the levator palpebrae superioris to keep my eyelids open, and that will also synapse directly on the muscle.”

Dominic considered the more complex answers, deciding on Abigail’s because of creativity. Abigail received Two points for the round, making her the WINNER of the game.

These are of course just examples of ways players could respond to a given prompt for the different cranial nerves. Depending on the number of players and rounds, as well as the individual creativity in their justifications, there are nearly limitless possibilities for how students can respond. I find that this encourages students to recall less obvious or discuss innervations of the cranial nerves and give the game plenty of “replay-ability” value.



Cranial Chaos